Student Information

Information	
Name	RENNA M. ISRAEL
Section	A
Assigned Dataset	Butterfly Image Classification
Kaggle Dataset URL	https://www.kaggle.com/datasets/phucthaiv02/butterfly-image-classification
Date Submitted	August 28, 2026

Part 1 — Dataset Exploration

1.1 Dataset Description

The Butterfly Image Classification is an assigned dataset. It is a set of images of various butterflies (or butterfly species) or classes. The labels for each image have also been provided. The dataset was analysed for the number of classes, images, image format, dimensions, channels and distribution of classes. A simple neural network was used to determine whether the butterfly images could be recognized and classified by the model or not.

1.2 Dataset Characteristics

Dataset Characteristic	Result
Number of Classes	75
Total Number of Images	6499
Image Format	JPEG
Image Dimensions	224 x 224
Color Channels	3
Smallest Class	WOOD SATYR (71 images)
Largest Class	MOURNING CLOCK (131 images)

1.3 Classes

#	Class Name	Number of Images
1	ADONIS	88
2	AFRICAN GIANT SWALLOWTAIL	75
3	AMERICAN SNOOT	74
4	AN 88	85
5	APPOLLO	90
6	ATALA	100
7	BANDED ORANGE HELICONIAN	97
8	BANDED PEACOCK	83
9	BECKERS WHITE	81

10	BLACK HAIRSTREAK	85
11	BLUE MORPHO	75
12	BLUE SPOTTED CROW	86
13	BROWN SIPROETA	99
14	CABBAGE WHITE	90
15	CAIRNS BIRDWING	83
16	CHECQUERED SKIPPER	95
17	CHESTNUT	85
18	CLEOPATRA	93
19	CLODIUS PARNASSIAN	87
20	CLOUDED SULPHUR	92
21	COMMON BANDED AWL	87
22	COMMON WOOD-NYMPH	90
23	COPPER TAIL	94
24	CRECENT	97
25	CRIMSON PATCH	72
26	DANAID EGGFLY	94
27	EASTERN COMA	93
28	EASTERN DAPPLE WHITE	92
29	EASTERN PINE ELFIN	95
30	ELBOWED PIERROT	82
31	GOLD BANDED	73
32	GREAT EGGFLY	78
33	GREAT JAY	94
34	GREEN CELLED CATTLEHEART	88
35	GREY HAIRSTREAK	86
36	INDRA SWALLOW	81
37	IPHICLUS SISTER	95
38	JULIA	81
39	LARGE MARBLE	81
40	MALACHITE	73
41	MANGROVE SKIPPER	87
42	MESTRA	86
43	METALMARK	76
44	MILBERTS TORTOISESHELL	96
45	MONARCH	90
46	MOURNING CLOAK	131
47	ORANGE OAKLEAF	87
48	ORANGE TIP	96
49	ORCHARD SWALLOW	76
50	PAINTED LADY	78
51	PAPER KITE	90
52	PEACOCK	84
53	PINE WHITE	86
54	PIPEVINE SWALLOW	84
55	POPINJAY	85
56	PURPLE HAIRSTREAK	79

57	PURPLISH COPPER	92
58	QUESTION MARK	77
59	RED ADMIRAL	82
60	RED CRACKER	96
61	RED POSTMAN	89
62	RED SPOTTED PURPLE	86
63	SCARCE SWALLOW	97
64	SILVER SPOT SKIPPER	83
65	SLEEPY ORANGE	107
66	SOOTYWING	90
67	SOUTHERN DOGFACE	87
68	STRAITED QUEEN	87
69	TROPICAL LEAFWING	83
70	TWO BARRED FLASHER	76
71	ULYSES	84
72	VICEROY	81
73	WOOD SATYR	71
74	YELLOW SWALLOW TAIL	75
75	ZEBRA LONG WING	76

1.4 Initial Observation

Based on the initial exploration of the Butterfly Image Classification dataset, I observed that the dataset contains 75 different butterfly classes with a total of 6,499 images. The number of images is different for the different classes, i.e., some butterfly classes have more images than others. I also noticed a wide range of differences in the appearance of the butterflies, such as variations in color, wing design, shape, and characteristics. Some classes look alike and it might be difficult to simply distinguish them from a neural network. Overall, the collection contains a good variety of images, which are suitable for testing an image classification model, and the variations in the amount of images in each class and the visual similarity between classes can have an impact on the classification performance of the model.

Part 2 — Train, Validation, and Test Sets

2.1 Dataset Splitting

The data set was split into three sets: **70% for training, 15% for validation and 15% for testing**. To ensure the class distribution is conserved across the subsets, train_test_split function was used with stratification by the class labels. A random_state equal to 42 was chosen in order to be able to reproduce the same split.

Dataset Split

Dataset	**Number of Images**	**Percentage**
Training	4,549	70%
Validation	975	15%
Testing	975	15%
Total	6499	100%

2.2 Purpose of Each Dataset

Training Set:

The butterfly data is divided into training data that the neural network learns to classify the different butterfly classes. The model consumes patterns and features from the training images, and updates the weights during training.

Validation Set:

The validation set is used to monitor the model's performance on images that were not used to directly train the model. It can be used to assess the model's performance in learning and to check for the possibility of overfitting.

Testing Set:

The testing set is the set of data used to test the trained model to complete the evaluation. It has images that were not used for training and validation, so that the capacity of the model to correctly classify new images of butterflies can be measured.

2.3 Observation

It's crucial that the test set is separate from the training set because the test set should be used to fairly assess the trained model's performance on images it's never been exposed to. In the case that identical pictures were used for training and testing, the model may just learn the training samples and obtain an unrealistic accuracy. The separation of the test set gives a more accurate assessment of the model's generalization capabilities to new images of butterflies.

Part 3 — Simple Neural Network

3.1 Model Architecture

Butterfly image classification has been performed by a simple feedforward structure neural network. The input images have a size of 32 × 32 pixels with 3 color channels (RGB). The Flatten layer creates a 1D vector of 3,072 elements (32 x 32 x 3), representing each image. Then a Dense layer with 128 neurons is added that learns patterns in the flattened image features. Lastly, the output Dense layer has 75 neurons, which represent the 75 butterfly classes in the data set. The output layer generates the classification output for the butterfly classes.

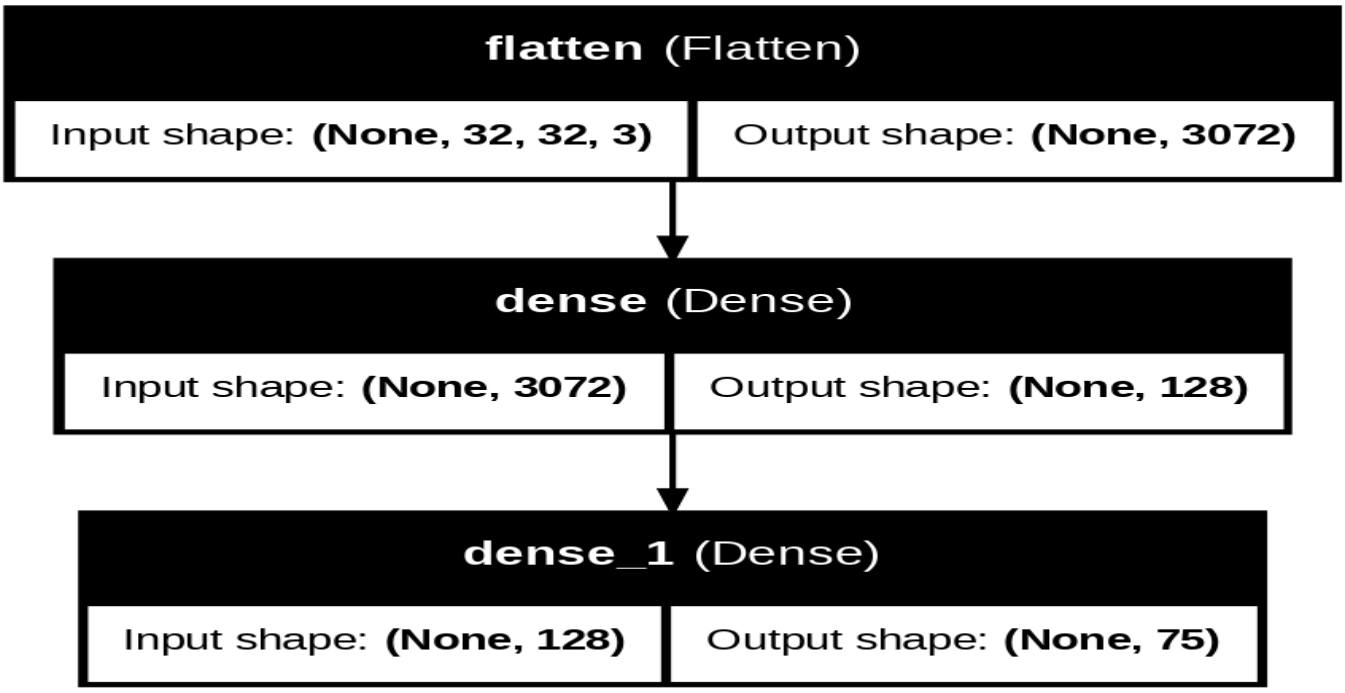


Figure 1: Architectural Diagram

3.2 Model Configuration

Component	Configuration
Input Image Size	32 x 32 x 3
Flatten	3, 072
Dense Units	128
Activation Function	relu/softmax
Optimizer	Adam
Loss Function	sparse_categorical_crossentropy
Number of Epochs	10
Batch Size	32

3.3 Understanding Flatten

Explain what happens when an image passes through the Flatten layer.

- When an image passes through the Flatten layer, the image's multidimensional pixel values are converted into a single one-dimensional vector. The image you will use for this activity is 32 x 32 x 3; the third number is the number of RGB colour channels. When the Flatten layer is applied to these values, 3,072 features are created, which is equal to 32 × 32 × 3. This enables passing the pixel information directly to the Dense layer of the simple neural network. The lab instructions explicitly state that Flatten has to be used to flatten each image to a 1D vector before it goes into Dense.

Part 4 — Model Training

4.1 Training Configuration

Parameter	Value
Epochs	10
Batch Size	32
Optimizer	Adam
Learning Rate	0.0010000000474974513
Training Images	4,549
Validation Images	975

4.1 Training and Validation Curves

Training and Validation Loss

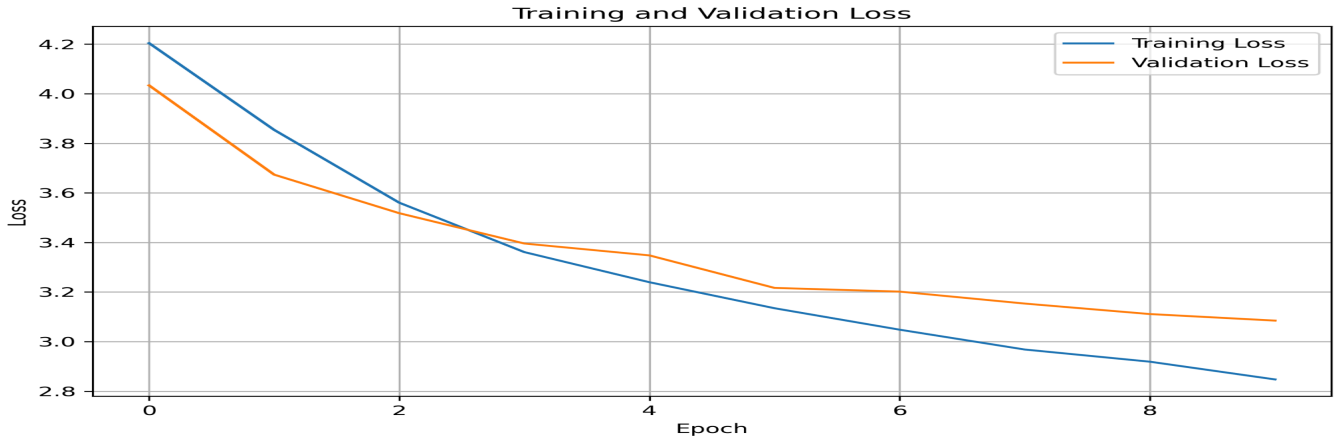


Figure 2. Training and Validation Loss

Training and Validation Accuracy

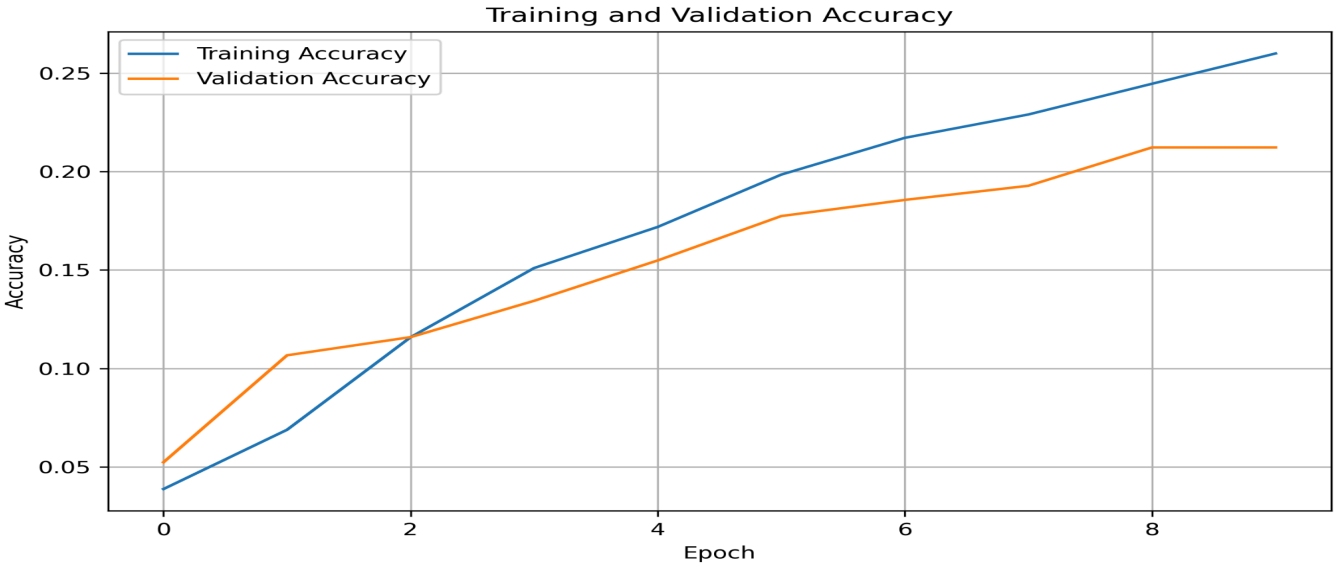


Figure 3. Training and Validation Accuracy

4.2 Interpretation

What do the training curves tell you about your model?

- The training and validation curves show how the model's performance changes as the number of epochs increases. In general, the loss during the training decreases and the training accuracy increases, meaning that the neural network is learning the patterns from the images of butterflies. Validation curves are provided to demonstrate the performance of the model on the validation set of images it has never seen before. In general, the closer the training and validation curves are together, the less severe the overfitting of the model. If, however, the training accuracy is increasing while the validation accuracy is not increasing or is decreasing, it may signify overfitting. In general, one can see if the model is learning and what the performance is in generalizing to images that it has not seen during the training process.

Part 5 — Experiment: Image Resolution

5.1 Experiment Objective

The objective of this experiment is to determine:

- The objective of this experiment is to determine **whether image resolution affects the performance of a simple neural network**. The same dataset and model will be used while changing only the image resolution to **32 × 32, 64 × 64, and 128 × 128 pixels**. The performance of each experiment will then be compared based on factors such as accuracy and training time.

5.2 Experimental Conditions

Experiment	Image Resolution	Input Features
Experiment 1	32 × 32 × 3	3,702
Experiment 2	64 × 64 × 3	12, 288
Experiment 3	128 × 128 × 3	49, 152

Part 6 — Experiment Results

6.1 Performance Comparison

Resolution	Input Features	Parameters	Training Accuracy	Validation Accuracy	Test Accuracy	Training Time
32 × 32	3072	403019	0.260057	0.212308	0.195897	15.136574
64 × 64	12288	1582667	0.073423	0.078974	0.066667	42.505402
128 × 128	49152	6301259	0.020224	0.020513	0.019487	171.601400

6.2 Accuracy Comparison

Test Accuracy by Image Resolution

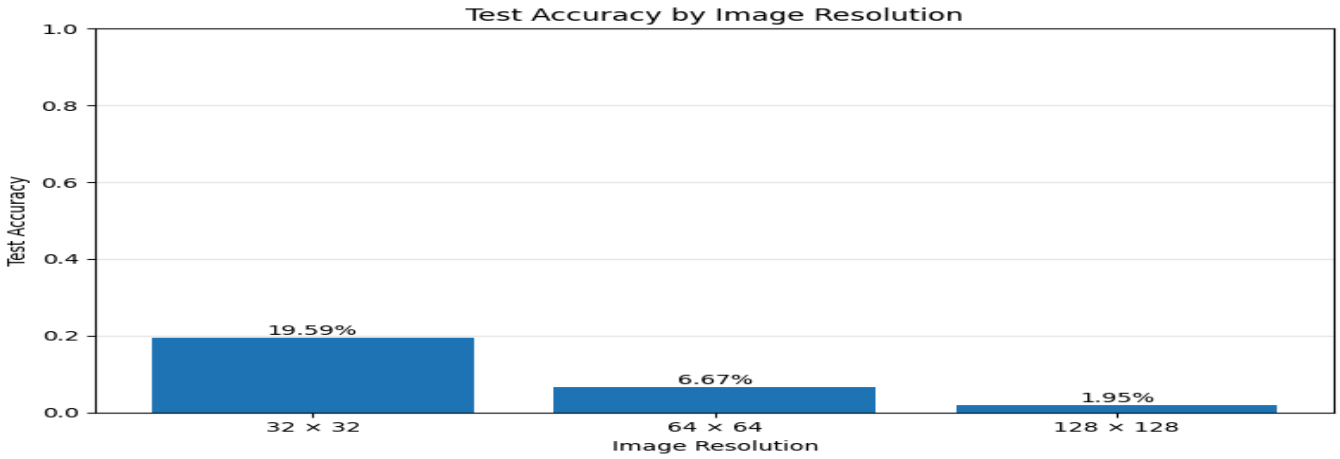


Figure 4. Test Accuracy by Image Resolution

6.3 Training Time Comparison

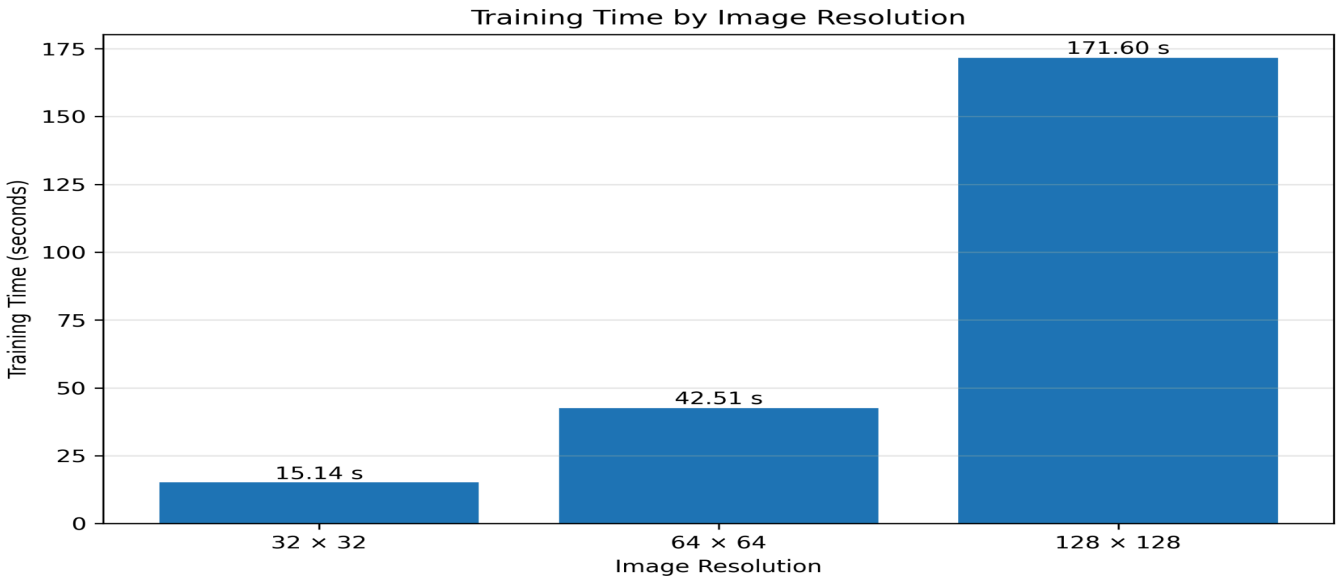


Figure 5: Training Time by Image Resolution

6.4 Experimental Observation

Describe what happened when the image resolution increased.

- As the image resolution increased from 32 × 32 to 64 × 64 and 128 × 128, the number of input features increased from 3,072 to 12,288 and 49,152, respectively. The higher the resolution, the more computational resources were needed, which means that the training time was also significantly increased. Based on the experiments, the training time increased from 15.14 seconds for 32 × 32 images to 42.51 seconds for 64 × 64 and 171.60 seconds for 128 × 128. The above illustrates that the higher the resolution of the images, the more information there is in the images, but the higher the cost of the model training.

Part 7 — Model Evaluation

7.1 Test Performance

Metric	Result
Test Loss	3.0943548679351807
Test Accuracy	0.1958974301815033

7.2 Confusion Matrix

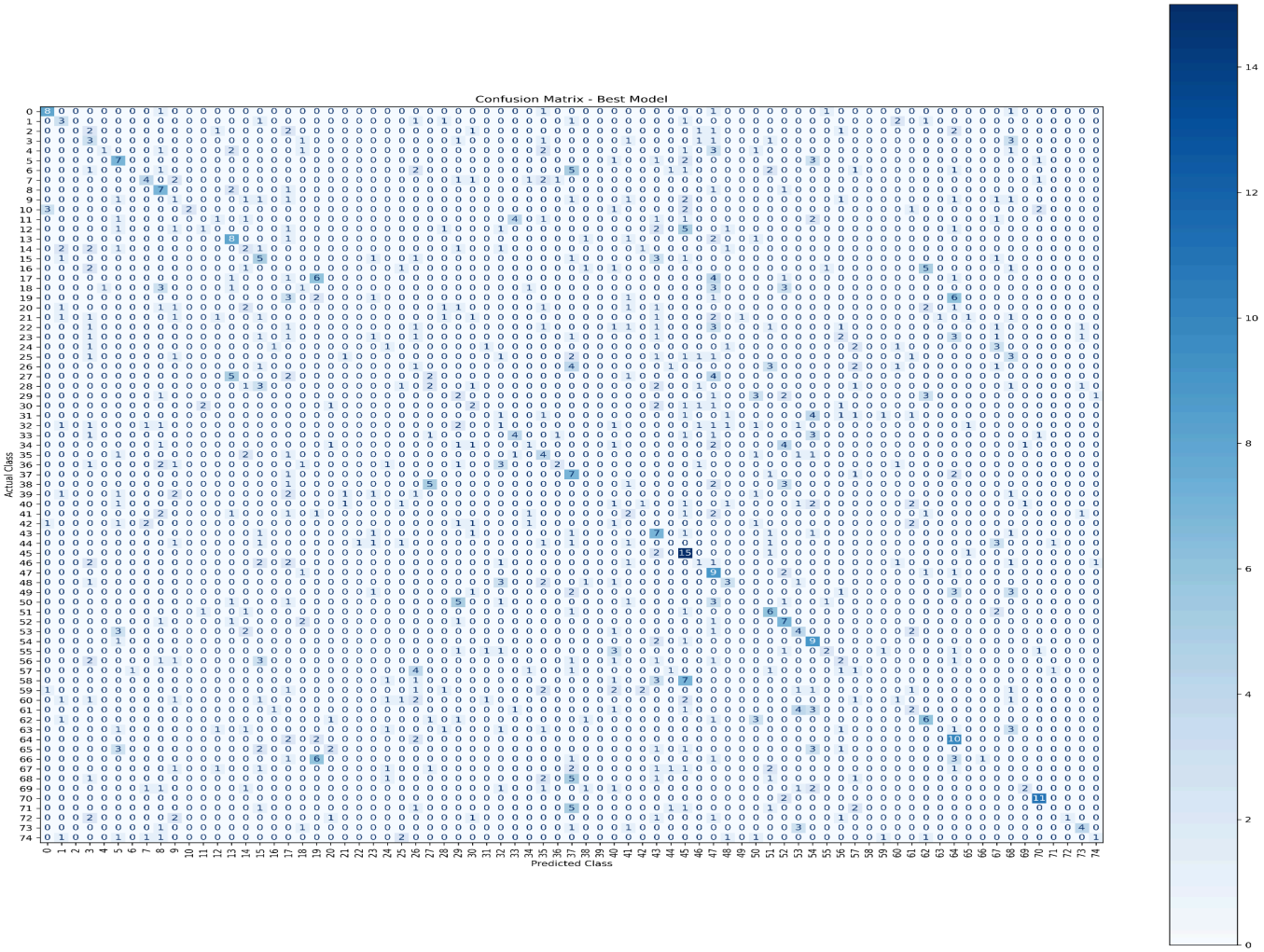


Figure 6. Confusion Matrix

7.3 Classification Performance

Class	Precision	Recall	F1-Score	Support
ADONIS	0.6154	0.6154	0.6154	13
AFRICAN GIANT SWALLOWTAIL	0.2308	0.2727	0.2500	11
AMERICAN SNOOT	0.0000	0.0000	0.0000	11
AN 88	0.1111	0.2308	0.1500	13
APPOLLO	0.5000	0.0769	0.1333	13
ATALA	0.2917	0.4667	0.3590	15
BANDED ORANGE HELICONIAN	0.0000	0.0000	0.0000	15

BANDED PEACOCK	0.4444	0.3077	0.3636	13
BECKERS WHITE	0.2692	0.5833	0.3684	12
BLACK HAIRSTREAK	0.0625	0.0769	0.0690	13
BLUE MORPHO	1.0000	0.1818	0.3077	11
BLUE SPOTTED CROW	0.0000	0.0000	0.0000	13
BROWN SIPROETA	0.0000	0.0000	0.0000	15
CABBAGE WHITE	0.3636	0.5714	0.4444	14
CAIRNS BIRDWING	0.1333	0.1667	0.1481	12
CHECQUERED SKIPPER	0.1923	0.3571	0.2500	14
CHESTNUT	0.0000	0.0000	0.0000	13
CLEOPATRA	0.0370	0.0714	0.0488	14
CLODIUS PARNASSIAN	0.1250	0.0769	0.0952	13
CLOUDED SULPHUR	0.1176	0.1429	0.1290	14
COMMON BANDED AWL	0.0000	0.0000	0.0000	13
COMMON WOOD-NYMPH	0.0000	0.0000	0.0000	14
COPPER TAIL	0.0000	0.0000	0.0000	14
CRECENT	0.1429	0.0714	0.0952	14
CRIMSON PATCH	0.1429	0.0909	0.1111	11
DANAID EGGFLY	0.0000	0.0000	0.0000	14
EASTERN COMA	0.0526	0.0714	0.0606	14
EASTERN DAPPLE WHITE	0.1667	0.1429	0.1538	14
EASTERN PINE ELFIN	0.0000	0.0000	0.0000	14
ELBOWED PIERROT	0.1053	0.1538	0.1250	13
GOLD BANDED	0.1818	0.1818	0.1818	11
GREAT EGGFLY	0.0000	0.0000	0.0000	12
GREAT JAY	0.0625	0.0714	0.0667	14
GREEN CELLED	0.4000	0.3077	0.3478	13

CATTLEHEART				
GREY HAIRSTREAK	0.1667	0.0769	0.1053	13
INDRA SWALLOW	0.1739	0.3333	0.2286	12
IPHICLUS SISTER	0.5000	0.1429	0.2222	14
JULIA	0.1628	0.5833	0.2545	12
LARGE MARBLE	0.0000	0.0000	0.0000	12
MALACHITE	0.0000	0.0000	0.0000	11
MANGROVE SKIPPER	0.0526	0.0769	0.0625	13
MESTRA	0.1429	0.1538	0.1481	13
METALMARK	0.0000	0.0000	0.0000	11
MILBERTS TORTOISESHELL	0.2000	0.4667	0.2800	15
MONARCH	0.0000	0.0000	0.0000	13
MOURNING CLOAK	0.2830	0.7895	0.4167	19
ORANGE OAKLEAF	0.1250	0.0769	0.0952	13
ORANGE TIP	0.1607	0.6429	0.2571	14
ORCHARD SWALLOW	0.3000	0.2500	0.2727	12
PAINTED LADY	0.0000	0.0000	0.0000	11
PAPER KITE	0.0000	0.0000	0.0000	14
PEACOCK	0.2727	0.5000	0.3529	12
PINE WHITE	0.2593	0.5385	0.3500	13
PIPEVINE SWALLOW	0.2353	0.3077	0.2667	13
POPINJAY	0.2647	0.6923	0.3830	13
PURPLE HAIRSTREAK	0.4000	0.1667	0.2353	12
PURPLISH COPPER	0.1429	0.1429	0.1429	14
QUESTION MARK	0.0769	0.0833	0.0800	12
RED ADMIRAL	0.0000	0.0000	0.0000	13
RED CRACKER	0.0000	0.0000	0.0000	14
RED POSTMAN	0.1429	0.0714	0.0952	14
RED SPOTTED PURPLE	0.1667	0.1538	0.1600	13

SCARCE SWALLOW	0.3000	0.4000	0.3429	15
SILVER SPOT SKIPPER	0.0000	0.0000	0.0000	12
SLEEPY ORANGE	0.2564	0.6250	0.3636	16
SOOTYWING	0.0000	0.0000	0.0000	13
SOUTHERN DOGFACE	1.0000	0.0769	0.1429	13
STRAITED QUEEN	0.0000	0.0000	0.0000	13
TROPICAL LEAFWING	0.0000	0.0000	0.0000	12
TWO BARRED FLASHER	0.5000	0.1667	0.2500	12
ULYSES	0.6471	0.8462	0.7333	13
VICEROY	0.0000	0.0000	0.0000	12
WOOD SATYR	1.0000	0.1000	0.1818	10
YELLOW SWALLOW TAIL	0.5000	0.3636	0.4211	11
ZEBRA LONG WING	0.3333	0.0909	0.1429	11

7.4 Class Performance
Best Performing Class

Class: ULYSES

Reason/Observation: **ULYSES** achieved the highest F1-score of **0.7333** among the 75 butterfly classes. This indicates that the model performed relatively well in correctly identifying ULYSES while maintaining a good balance between precision and recall.

Most Difficult Class

Class: AMERICAN SNOOT

Reason/Observation: **AMERICAN SNOOT** had the lowest F1-score of **0.0000**, indicating that the model had the greatest difficulty correctly identifying this class. This suggests that its images may have been frequently misclassified as other butterfly classes.

Part 8 — Prediction Analysis

8.1 Correct Predictions

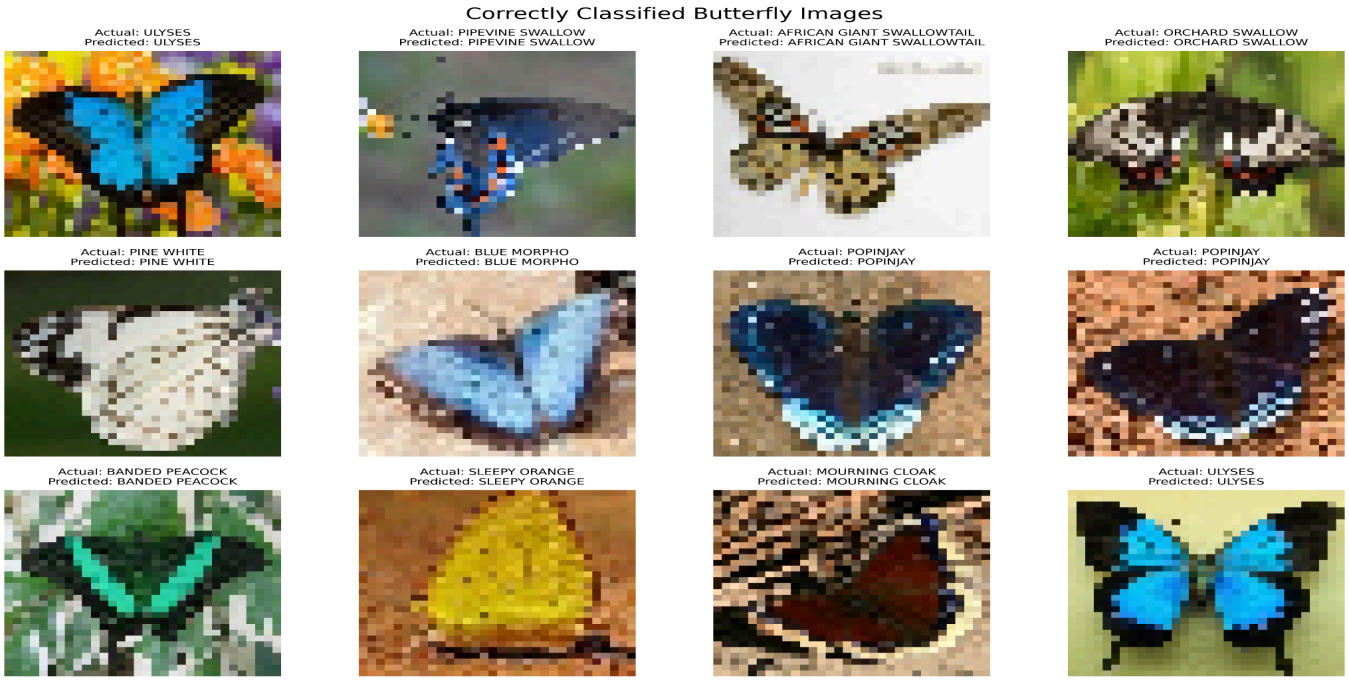


Figure 7. Correct Predictions

The correctly classified images generally have visual features that the model was able to recognize and associate with the appropriate butterfly class. These images may have relatively clear butterfly shapes, visible wing patterns, and distinguishable colors or markings. The correct predictions indicate that the model was able to learn some of the visual characteristics used to differentiate between butterfly classes.

8.2 Incorrect Predictions

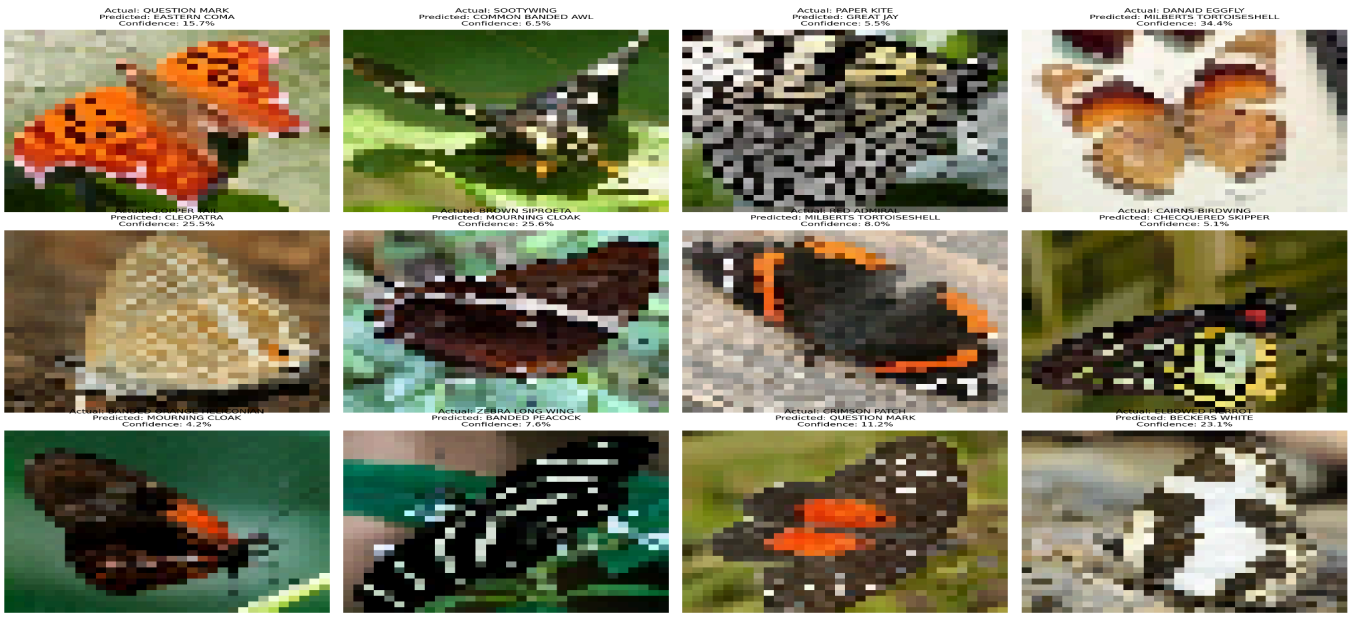


Figure 8. Incorrect Predictions

8.3 Prediction Errors

Choose at least **three incorrect predictions** and analyze them.



Figure 9: Prediction Errors

Image	Actual Class	Predicted Class	Possible Reason
1	QUESTION MARK	EASTERN COMA	The model may have difficulty distinguishing visually similar butterfly classes based on shape, wing patterns, colors, or other visual features.
2	SOOTYWING	COMMON BANDED AWL	The model may have difficulty distinguishing visually similar butterfly classes based on shape, wing patterns, colors, or other visual features.
3	PAPER KITE	GREAT JAY	The model may have difficulty distinguishing visually similar butterfly classes based on shape, wing patterns, colors, or other visual features.

Discussion

Why do you think the neural network confused these classes?
The neural network may have confused these classes because some butterfly species have similar visual characteristics, such as wing color, patterns, shape, and markings. Differences in image background, butterfly orientation, lighting, and image quality may also make classification more difficult. Since the model uses a simple neural network, it may not capture complex visual features as effectively as more advanced convolutional neural networks.

Part 9 — Reflection

9.1 What Did You Observe?

I observed that the image resolution had a noticeable effect on the performance and training time of the neural network. As the resolution increased from **32 × 32** to **64 × 64** and **128 × 128**, the number of input features increased significantly, which also caused the training time to increase. I also observed that the model was able to correctly classify some butterfly classes, such as **ULYSES**, but had difficulty with others, such as **AMERICAN SNOOT**. Overall, the experiment showed that higher image resolution provides more image information but also requires greater computational resources.

Part 10 — Conclusion

The dataset used in this experiment was a butterfly image dataset containing **75 classes and 6,499 images**. The simple neural network was able to recognize and classify butterfly images, although its performance varied across different classes. **ULYSES** was the best-performing class with an F1-score of **0.7333**, while **AMERICAN SNOOT** was the most difficult class, with an F1-score of **0.0000**. Increasing the image resolution from **32 × 32** to **64 × 64** and **128 × 128** increased the number of input features and significantly increased training time, from **15.14 seconds** to **42.51 seconds** and **171.60 seconds**, respectively. The major limitations encountered included difficulty distinguishing visually similar butterfly classes and the increased computational cost of processing higher-resolution images. Overall, the experiment shows that **a simple neural network can recognize images**, but its ability to accurately distinguish many visually similar classes is limited compared with more advanced image-classification models.

Part 11 — Appendix

Appendix A — Google Colab Notebook

Colab Link:

[https://colab.research.google.com/drive/1-i6atgJGsZjcRZF_L4hyrzGVnTgvcxpL#scrollTo=fCBDf2BBzULo]

Appendix B — Additional Results

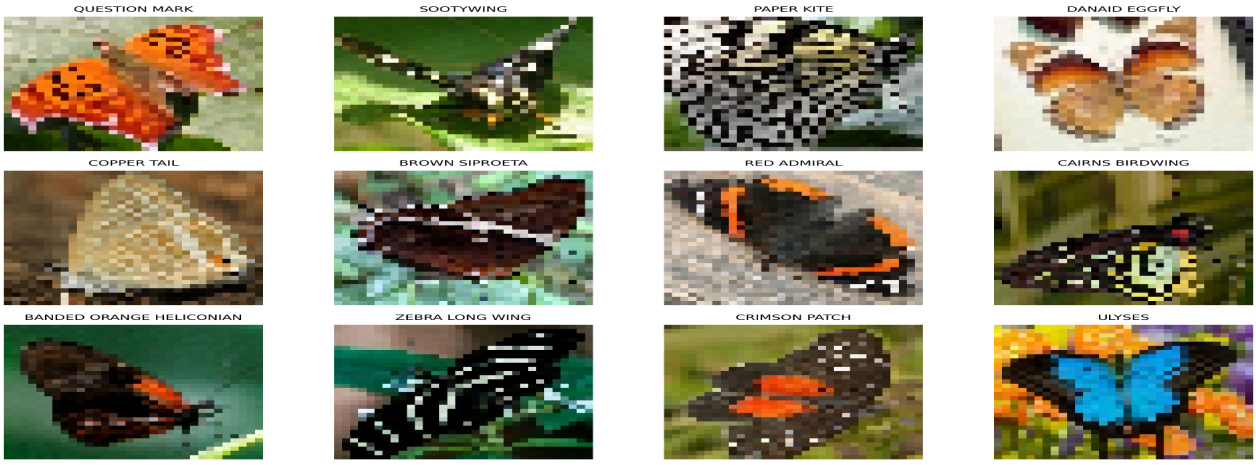


Figure B.1. Sample Images from the Butterfly Dataset

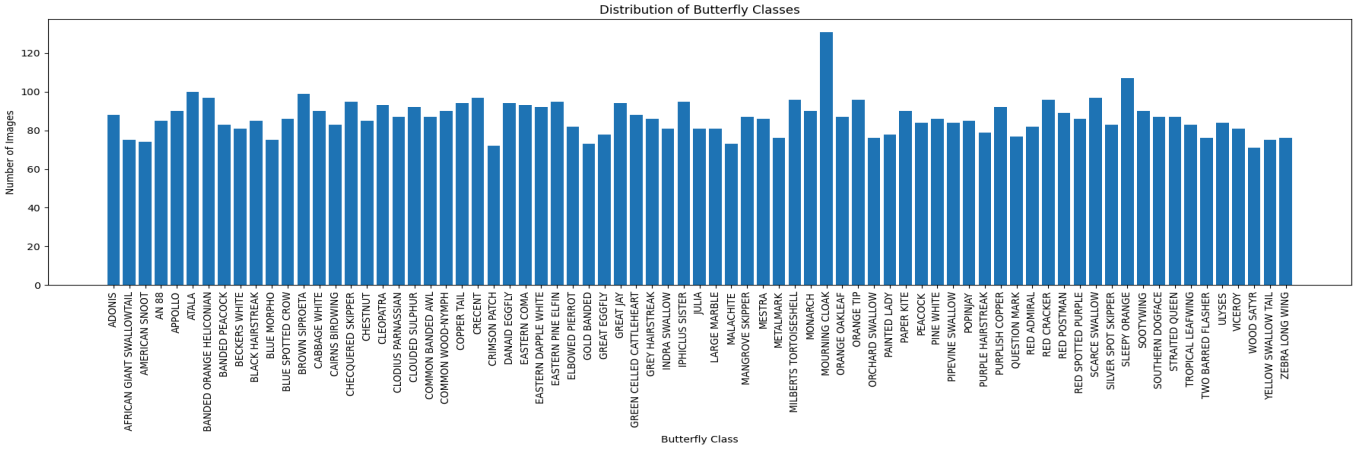


Figure B.2. Distribution of Images Across the 75 Butterfly Classes